

Note – Ensure date and time are visible.

1. Download and install OSQUERY on a fresh machine (either revert if snapshot is present or install a new CentOS) and display the following details using appropriate queries.
 - a. System inventory
 - b. Logged-in user information
 - c. Running processes
 - d. Installed software
 - e. Open network connections
2. Collect the logs for the following and verify/list them.
 - a. Domain firewall
 - b. Public firewall
 - c. Private firewall
 - d. Linux logs
 - e. Save Linux firewall logs to a file “/tmp/firewalld.log”
3. Perform RanSIM simulation by installing and configuring it on a windows server DC machine.
4. Install and configure Infection Monkey and scan possible VMs within the environment and search for vulnerabilities.
5. Write powershell script for the following:
 - a. Last 2 failed log-ons events
 - b. Last 2 successful logon events
 - c. Search for encoded command within logs.
 - d. Search for all the logs for administrator.
 - e. Windows Defender logs
6. Perform the following:
 - a. Use cmdlet “Get-service” and print Base64 encoded value of it and run it using powershell command.
 - b. Use the encoded string and decode it and display equivalent cmdlet by executing it.
“VwByAGkAdABIAC0ASABvAHMAdAAgACcAVwBIAgWAYwBvAG0AZQAgAHQAbwAgAFAAcgBvAGoAZQBjAHQAIAA1ACcA”.
7. Add the following port numbers/services and validate using commands on windows and linux machines with NO ERRORS:
 - a. Add port 443
 - b. Add port 53
 - c. Add Samba service
 - d. Block ICMP ping request.
 - e. Block NTP ports
8. List top 10 system event logs from DC machine and save them into a text file (created using cmdlet).
9. Add a 20GB HDD on this domain controller machine with NTFS file system. Then extend 5 GB more to the existing one and verify.
10. Use powershell cmdlet to list open ports in DC machine and ensure to only display IPV4 based ports.